

**GUYANA
NATIONAL GRADE SIX ASSESSMENT**

ANSWER BOOKLET

**FILL IN ALL THE INFORMATION THAT IS REQUIRED CLEARLY
AND LEGIBLY IN CAPITAL LETTERS.**

TEST CODE

0	9	6	3	4	0	2	0
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SUBJECT MATHEMATICS – Paper 02

WRITE YOUR CANDIDATE NUMBER HERE.

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**GUYANA
NATIONAL GRADE SIX ASSESSMENT**

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For CXC® Use only

Question Numbers	Profiles			Total	Initials
	K	AT	R		
1					
2					
3					
4					
5					
6					
Total					

**WRITE THE NAME OF
YOUR SCHOOL BELOW.**

WRITE YOUR FULL NAME BELOW.

DATE OF BIRTH

<i>Day</i>	<i>Month</i>	<i>Year</i>
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SEX MALE ☐
FEMALE ☐

DO NOT WRITE IN THIS SPACE.

FOLDER NUMBER

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DO NOT WRITE IN THIS SPACE.

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**DO NOT
WRITE ON
THIS PAGE**

TEST CODE 09634020

FORM TP 2021388

GUYANA
MINISTRY OF EDUCATION
NATIONAL GRADE SIX ASSESSMENT

MATHEMATICS

PAPER 02

AUGUST 5, 2021 (a.m.)

Reading Time: 10 minutes

Writing Time: 60 minutes

Read these instructions carefully before you attempt to answer the questions.

1. Write your candidate number clearly on each page.
2. This paper contains **six** questions. You are required to answer **Question 1** and **three** others. **Each** question is worth **5** marks.

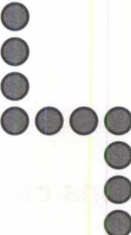
Note: You must answer **only four** questions. Be sure to answer **all** parts of the **four** questions fully.

3. You have 60 minutes of writing time and 10 minutes of reading time. Writing may begin during the reading time.
4. Write the answer for each question in the space provided in this booklet.
5. Answers **must** be written in complete sentences where possible.
6. Ensure that your handwriting is clear.
7. If you have to erase, do so cleanly.
8. Each step of your work **must** be **clearly** shown in this booklet.
9. Look over your work when you have finished.
10. Do **not** take away any part of this booklet.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

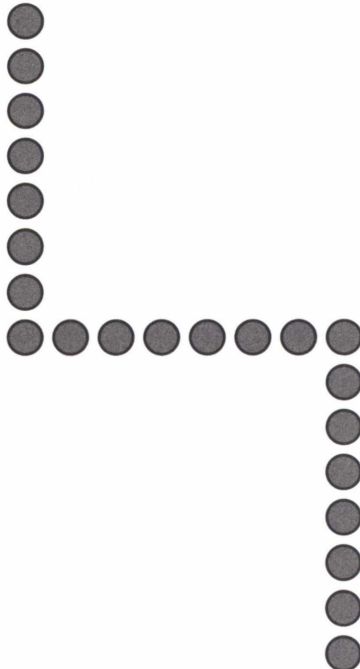
Candidate number:

Instruction: Question 1 is COMPULSORY. You MUST answer it.

No.	TEST ITEMS
1.	(a) Zara used some buttons to make a sequence of figures which form a pattern. The first 4 figures in the sequence are shown below. Study the pattern and answer the questions that follow.     Figure 1 Figure 2 Figure 3 Figure 4 (i) Draw Figure 5 of the sequence. (1 mark)
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Candidate number:

No.	TEST ITEMS
	(ii) Which figure is shown in the diagram below?  Answer: Figure _____ (1 mark) (iii) How many buttons will be in Figure 10? Answer: _____ buttons (1 mark)
DO NOT WRITE HERE	K AT R

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Candidate number:

No.	TEST ITEMS	WORKING COLUMN
	(b) Zaheer had a set of marbles which he used to make a design. He used $\frac{2}{3}$ of the number of marbles and had 14 left. How many marbles did he use to make the design? Answer: _____ marbles (2 marks) Total 5 marks	
DO NOT WRITE HERE		K (1)
		AT (2)
		R (2)
		TOTAL (5)

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Candidate number:

Instruction: Answer any three questions from Questions 2 to 6.

No.	TEST ITEMS	WORKING COLUMN
2.	(a) Look at the flowchart below. <pre> graph TD A[10] --> B((+ 2)) B --> C[12] C --> D((÷ 3)) D --> E[4] </pre> Here is another flowchart. Fill in the 4 missing numbers. <pre> graph TD A[] --> B((+ 2)) B --> C[] C --> D((÷ 3)) D --> E[25] E --> F((+ 2)) F --> G[] G --> H((÷ 3)) H --> I[] </pre> (2 marks)	
DO NOT WRITE HERE K AT R		

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Candidate number:

No.	TEST ITEMS	WORKING COLUMN
	(b) The elements of two sets, A and B, are listed below. Set A = {Factors of 8} Set B = {Factors of 16} List the elements of (i) Sets A and B Set A = { _____ } Set B = { _____ } (1 mark) (ii) the set that is made up of factors of both 8 and 16. Answer: { _____ } (1 mark)	
DO NOT WRITE HERE	K	
AT		R

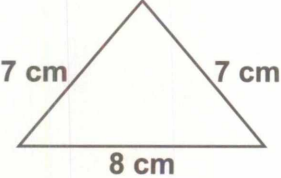
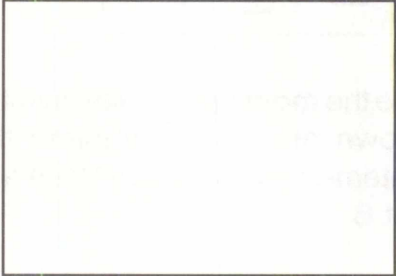
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Candidate number:

No.	TEST ITEMS	WORKING COLUMN									
	Look at the symbols below. $=$ $\in$ $\subset$ (iii) Use the most appropriate symbol shown above to complete the statement below about Set A and Set B. A _____ B (1 mark) Total 5 marks										
<table border="1" style="width: 100%;"><tr><td style="width: 33%;">DO NOT WRITE HERE</td><td style="width: 10%;">K (1)</td><td style="width: 10%;"></td><td style="width: 10%;">AT (2)</td><td style="width: 10%;"></td><td style="width: 10%;">R (2)</td><td style="width: 10%;"></td><td style="width: 10%;">TOTAL (5)</td><td style="width: 10%;"></td></tr></table>			DO NOT WRITE HERE	K (1)		AT (2)		R (2)		TOTAL (5)	
DO NOT WRITE HERE	K (1)		AT (2)		R (2)		TOTAL (5)				

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Candidate number:

No.	TEST ITEMS
3.	(a) In the diagrams below, the perimeter of the triangle is half the perimeter of the rectangle.  7 cm 7 cm 8 cm  9 cm (i) Calculate the perimeter of the rectangle. Answer: _____ cm (2 marks) (ii) Find the length of the rectangle. Answer: _____ cm (1 mark)
DO NOT WRITE HERE	K AT R

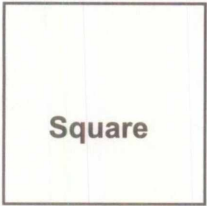
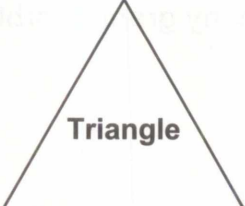
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Candidate number:

No.	TEST ITEMS									
	(b) A bag has 50 marbles of 3 different colours: red, blue and green. Half the number of marbles are red, 15 are blue and the remainder green. (i) How many green marbles are in the bag? Answer: _____ green marbles (1 mark) (ii) Express, as a ratio, the quantity of red to blue marbles. Answer: _____ (1 mark) Total 5 marks									
<table border="1"><tr><td>DO NOT WRITE HERE</td><td>K (1)</td><td></td><td>AT (2)</td><td></td><td>R (2)</td><td></td><td>TOTAL (5)</td><td></td></tr></table>		DO NOT WRITE HERE	K (1)		AT (2)		R (2)		TOTAL (5)	
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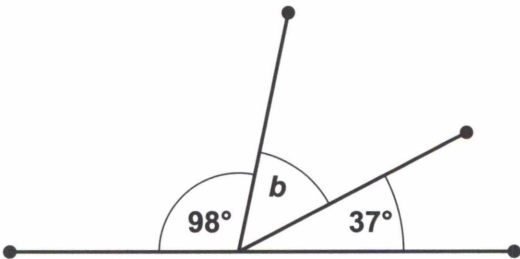
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Candidate number:

No.	TEST ITEMS
4.	(a) Look at the 4 polygons shown below.  Square  Pentagon  Triangle  Parallelogram (i) State one difference in property between the pentagon and the parallelogram. _____ _____ (1 mark) (ii) State the name of the shape that has both parallel and perpendicular sides. _____ _____ (1 mark) (iii) Which shape has only acute angles? _____ _____ (1 mark)
DO NOT WRITE HERE K AT R	

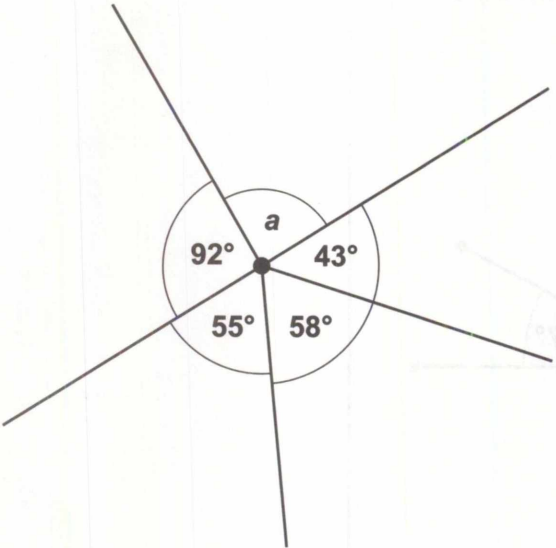
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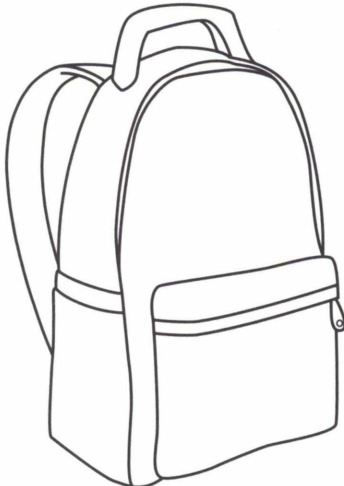
No.	TEST ITEMS	WORKING COLUMN
	(b) Find the value of the angles marked with a letter in each diagram below. (i)  Answer: $b =$ _____ degrees (1 mark)	
DO NOT WRITE HERE	K	
AT		R

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Candidate number:

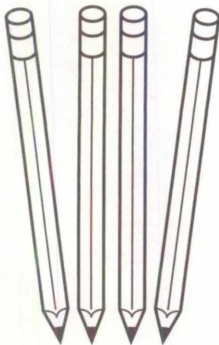
No.	TEST ITEMS	WORKING COLUMN
	(ii)  Answer: $a =$ _____ degrees (1 mark) Total 5 marks	
DO NOT WRITE HERE	K (1)	AT (2)
R (2)	TOTAL (5)	

Candidate number:

No.	TEST ITEMS	WORKING COLUMN
5.	(a) Kelly bought the haversack shown below for \$3 600.  Cost price \$3 600 She sells it to make a profit of \$720. (i) What is the selling price of the haversack? Answer: \$ _____ (1 mark) (ii) Express Kelly's profit as a percentage of the cost price of the haversack. Answer: _____ % (1 mark)	
DO NOT WRITE HERE		K AT R

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Candidate number:

No.	TEST ITEMS	WORKING COLUMN
	(b) Look at the advertisement of a sale shown below.   4 pencils for \$100  5 books for \$1 000 Mr Melville had \$4 000. He spent \$3 000 on books. (i) How many books did he buy? Answer: _____ books (1 mark)	
DO NOT WRITE HERE	K	AT
R		

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Candidate number:

No.	TEST ITEMS	WORKING COLUMN
	(ii) Mr Melville must buy 100 pencils. How much more money does he need to buy the pencils? Answer: \$ _____ (2 marks) Total 5 marks	
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		AT (2)
		R (2)
		TOTAL (5)

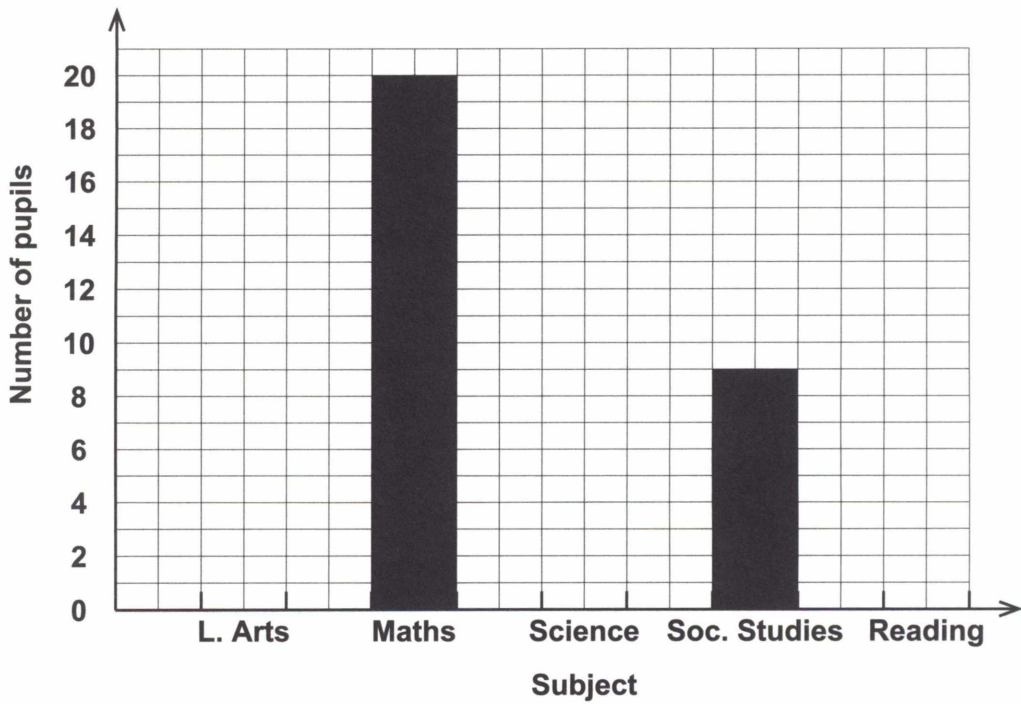
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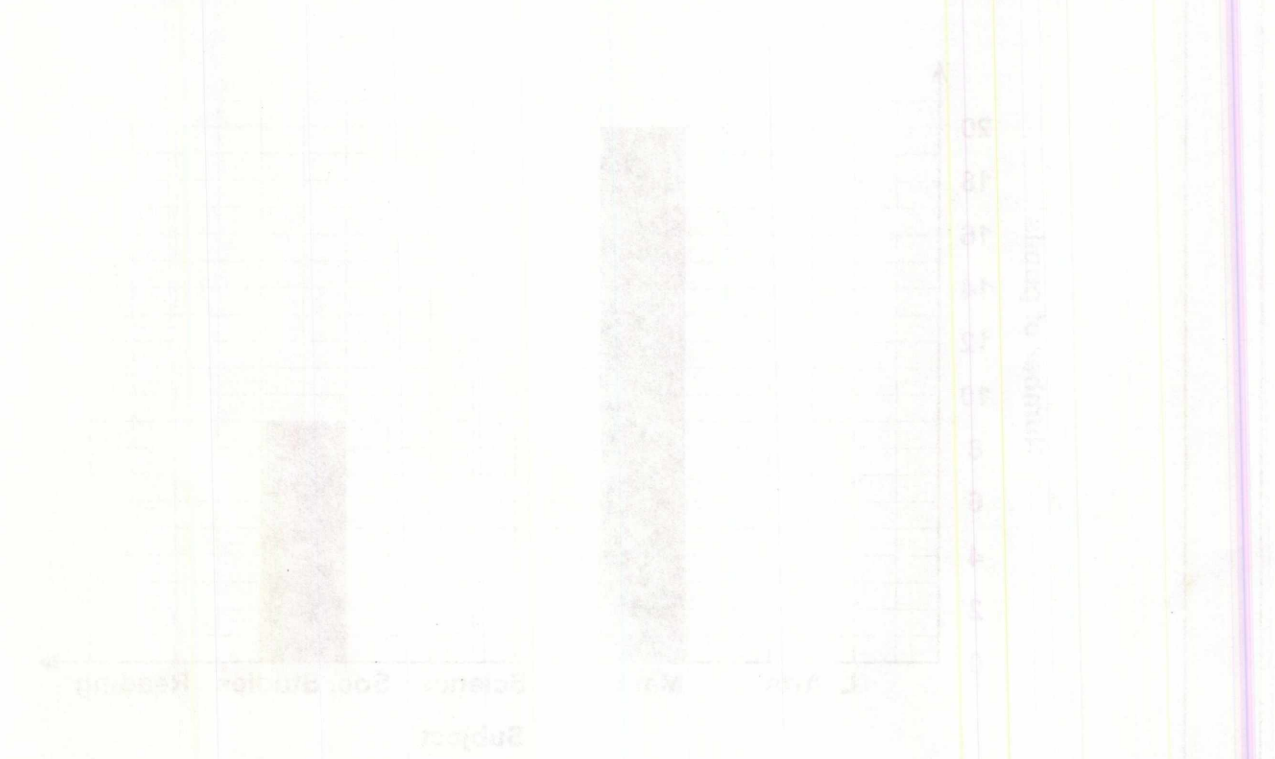
No.	TEST ITEMS												
6. The tally chart below shows the number of pupils who passed each of 5 subjects in the end of term examination. All of the 25 pupils in the class were present for each test. Study the chart carefully and then answer the questions that follow. <table border="1" style="margin: 20px auto; border-collapse: collapse; text-align: center;"> <thead> <tr> <th style="width: 30%;">Subject</th> <th>Tally</th> </tr> </thead> <tbody> <tr> <td>Language Arts</td> <td> </td> </tr> <tr> <td>Mathematics</td> <td> </td> </tr> <tr> <td>Science</td> <td> </td> </tr> <tr> <td>Social Studies</td> <td> </td> </tr> <tr> <td>Reading</td> <td> </td> </tr> </tbody> </table> (a) How many pupils passed Science? _____ pupils (1 mark) (b) Which two subjects did the same number of pupils pass? _____ and _____ (1 mark) (c) Which subject did the highest number of pupils not pass? _____ (1 mark)	Subject	Tally	Language Arts		Mathematics		Science		Social Studies		Reading		
Subject	Tally												
Language Arts													
Mathematics													
Science													
Social Studies													
Reading													
DO NOT WRITE HERE K AT R													

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Candidate number:

No.	TEST ITEMS																					
(d) Using the information from the tally chart on page 18, complete the bar graph below to show the number of pupils who passed each subject.  <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <caption>Data for Bar Graph</caption> <thead> <tr> <th>Subject</th> <th>Number of pupils</th> </tr> </thead> <tbody> <tr> <td>L. Arts</td> <td>0</td> </tr> <tr> <td>Maths</td> <td>20</td> </tr> <tr> <td>Science</td> <td>0</td> </tr> <tr> <td>Soc. Studies</td> <td>9</td> </tr> <tr> <td>Reading</td> <td>0</td> </tr> </tbody> </table> (2 marks) Total 5 marks END OF TEST IF YOU FINISH BEFORE TIME IS UP, CHECK YOUR WORK ON THIS TEST.	Subject	Number of pupils	L. Arts	0	Maths	20	Science	0	Soc. Studies	9	Reading	0	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 25%;">DO NOT WRITE HERE</td> <td style="width: 10%;">K (1)</td> <td style="width: 10%;"></td> <td style="width: 10%;">AT (2)</td> <td style="width: 10%;"></td> <td style="width: 10%;">R (2)</td> <td style="width: 10%;"></td> <td style="width: 10%;">TOTAL (5)</td> <td style="width: 10%;"></td> </tr> </table>	DO NOT WRITE HERE	K (1)		AT (2)		R (2)		TOTAL (5)	
Subject	Number of pupils																					
L. Arts	0																					
Maths	20																					
Science	0																					
Soc. Studies	9																					
Reading	0																					
DO NOT WRITE HERE	K (1)		AT (2)		R (2)		TOTAL (5)															

(b) Using the information on the table, complete the graph below to show the number of subjects who passed the subject.



END OF TEST

IF YOU FINISH BEFORE TIME, USE YOUR WORK ON THIS TEST

Prepared by
CARIBBEAN EXAMINATIONS COUNCIL